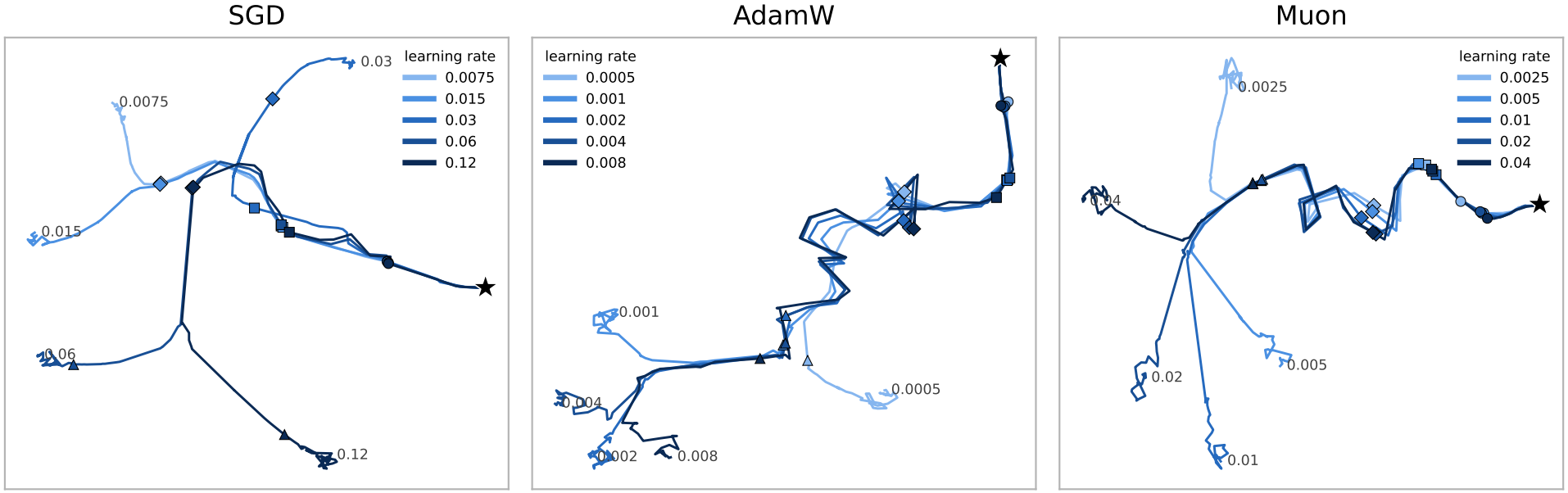
Setting: TinyStories-10M, as in trajectories_methods.pdf, data order d0 only, shared initialization; learning rates $\frac{1}{4}$, $\frac{1}{2}$, 1, 2, 4 × each optimizer's main rate: SGD 0.0075–0.12, AdamW 5×10^{-4}–8×10^{-3}, Muon 0.0025–0.04; 10,000 steps; the 1× run is the main run. densMAP of the square-root Jensen–Shannon distance, one embedding per optimizer.



Setting: TinyStories-10M, 10,000 steps, linear warmup 200 then cosine to 10%, data order 0, shared initialization. Sharpness: $\lambda_{\max}(H)$ for SGD and Muon, $\lambda_{\max}(P^{-1}H)$ for AdamW.

Solid: sharpness of each run; dashed, same colour: its stability threshold at the instantaneous rate $\eta(t) = \eta\,\varphi(t)$; grey band: plateau window (after warmup, $\varphi(t) \geq 0.8$).

